



DTM-CC016A Single Antenna Multi-Channel Transceiver Gateway Module

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单天线多通道收发网关模组

Overview

DTM-CC016A gateway module is a single-antenna multi-channel transceiver gateway module developed by Shanghai Panqi Microelectronics Co., Ltd. based on the PAN3029 chip. The module has a built-in M4 processor, integrated PA and LNA, and can efficiently process Chirp-IoT data transmission and reception. The module operates in the 470-510MHz frequency band and supports rate adaptation and communication data AES encryption.

Users can drive the module through the SPI interface. The gateway module supports RF data reception, data transmission, parameter configuration and other interfaces, which is convenient for users to use. The gateway module can be widely used in one-to-many star networks, supports large-capacity terminal node networks, supports ChirpLAN Mode A/B/C protocols, supports multiple application scenarios, etc.

Key Features

- Operating frequency band: 470~510MHz
- Modulation: Chirp-IoT
- Channels: 8RX+1TX, half-duplex
- Communication rate: Support 8 Chirp-IoT data channels, adaptive
- Receiving sensitivity: -142dBm@SF12 BW62.5kHz
- Maximum transmit power: 28dBm
- Encryption mode: AES hardware encryption
- Communication interface: SPI
- RSSI detection and output
- ChirpLAN Mode A/B/C protocols

Application Areas

- Smart city
- Smart rubber tapping
- Smart farm
- Smart power and other industrial scenarios



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1 Hardware

1.1 Physical Picture



Figure 1-1 Module physical picture





Figure 1-2 Gateway physical picture (Raspberry Pi + baseboard + module)

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1.3 Module dimensions

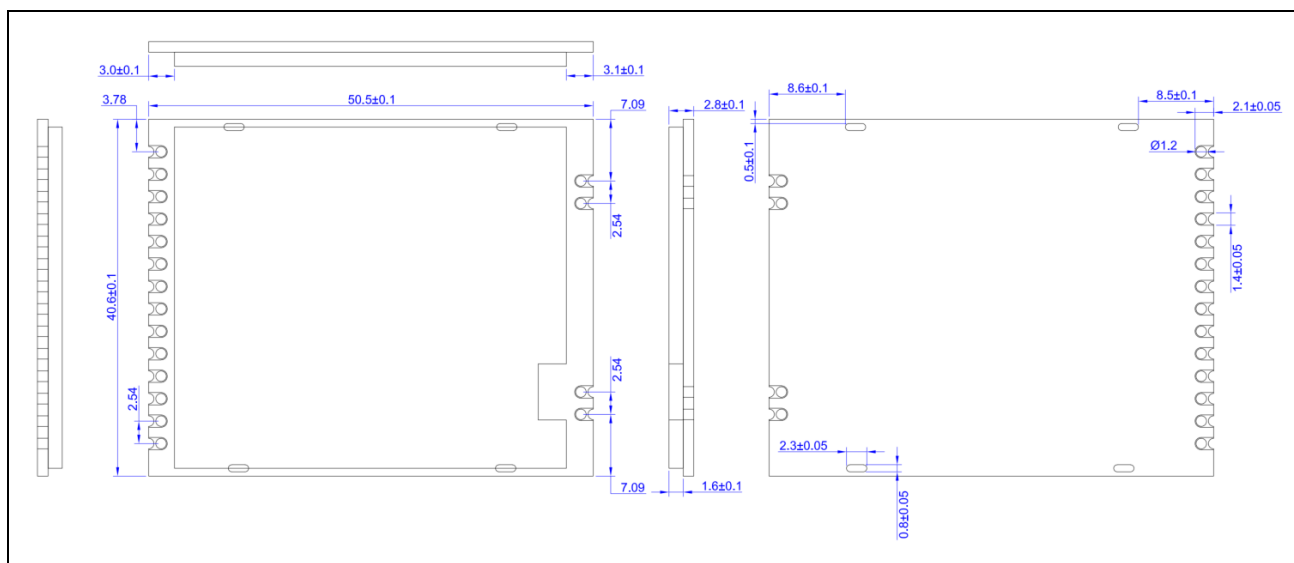


Figure 1-4 Gateway module dimensions

1.4 Module Pin Definition

Port	Description
SPI communication port	1 SPI CSN1_0, MOSI1, MISO1, maximum rate 10MHz
IO1	Custom function pin
Reset	Reset port, RST, 0 active
Programming port	SWD, SWCLK
Serial communication port	UART TX, RX
Power supply	VCC 1PIN, GND 2PIN
Power control	PWR_EN module power switch control pin; 0 - off, 1 - on

1.5 Module baseboard interface definition

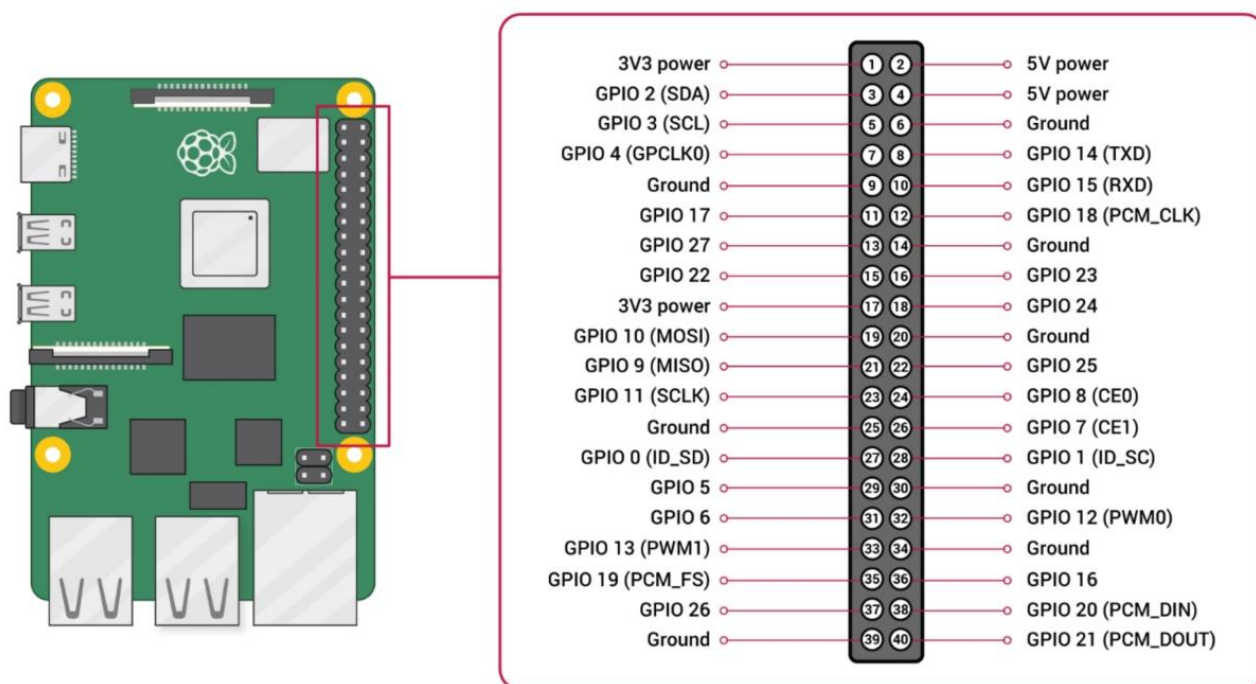


Figure 1-5 Gateway module baseboard pin diagram

1.6 Electrical parameters

Parameter	Description
Power supply voltage	Nominal voltage 5V/2A, input range (4.5V-18V)
Receive current	72mA
Transmit current	870mA@28dBm,470MHz
RF interface	50Ω

1.7 Physical properties

Parameter	Description
Size	50.5*40.6mm
Package	Stamp hole
Temperature	-40~85°C (operating), -55~105°C (storage)
Humidity	5~95% (no condensation)

1.8 RF Parameters

Parameter	Description
Output power	Maximum 28dBm
Bandwidth	62.5kHz, 125kHz, 250kHz, 500kHz
Spreading factor	SF5~SF12
Receiver sensitivity	-142dBm@SF12 BW62.5kHz
Frequency range	470~510MHz
Channels	8 ChirpLoT receive channels, 8RX+1TX half-duplex

2 Module Protocol

The gateway module provides an SPI interface to the outside world. The module acts as an SPI slave. Our company provides Raspberry Pi master control operation examples and gateway module programs. The open source address is: <https://gitee.com/chirplan>

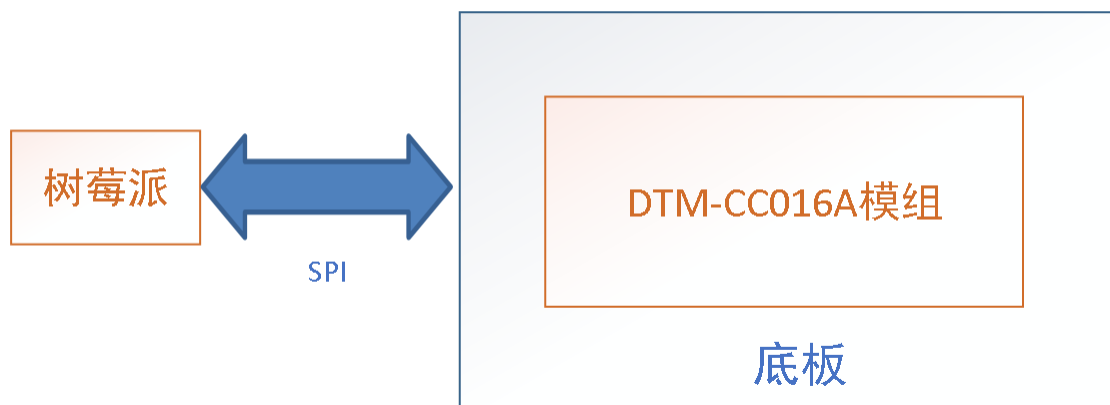


Figure 2-1 Gateway system schematic diagram

The SPI communication protocol between the module and the main control MCU (Raspberry Pi) is as follows:

(1) The Raspberry Pi obtains the RF data received by the module. The received data content format is as follows:

接收数据区	数据段起始地址	数据存储 (buffer size)	说明
区域说明	0	buf[0]接收区数据大小低8位	SPI主机读取. 从机判断等于0时将数据放置接收区, 然后将数据大小写至该区域
接收数据FIFO检查区	1	buf[1]接收区数据大小高8位	
接收数据FIFO数据区	2	0xxx	Packet数量(协议示例此值为2)
	3	payload_size	payload长度
	0+4	PAYLOAD	PAYLOAD DATA
	...		
	...		
	...		
	payload_size+3	Channel	信道号 (0-7)
	payload_size+4	SF[7:4]CR[3:1]CRC_EN[0]	扩频因子5-12, CR:1-4, CRC:0-1
	5+payload_size	BW[7:4]LDR[0]	带宽 (6\7\8\9分别代表62.5, 125, 250, 500) LDR0/1
	6+payload_size	RSSI	信号质量
	7+payload_size	payload_size	
		PAYLOAD	
		Channel	
		SF[7:4]CR[3:1]CRC_EN[0]	
		BW[7:4]LDR[0]	
		RSSI	
		CRC_L	CRC16低8位(上面所有字节)
		CRC_H	CRC16高8位

Buffer size

DTM-CC016A Single Antenna Multi-Channel Transceiver Gateway Module

(2) The master control needs to send data through the gateway module.
The protocol format is as follows:

发送数据区		
数据段起始地址	数据存储	说明
0	CMD_TX_DATA(0x0B)	发送数据命令
1	buffer size[0]	整个数据包的长度低字节
2	buffer size[1]	整个数据包的长度高字节
3	Channel	发送信道号（8-15）
4	SF[7:4]CR[3:1]CRC_EN[0]	扩频因子5-12, CR:1-4, CRC:0-1
5	BW[7:4]LDR[0]	带宽（6\7\8\9分别代表62.5, 125, 250, 500）LDR0/1
6	Power	发射功率
7	Preamble_L	前导码个数低8位
8	Preamble_M	前导码个数高8位
9	MAPM_Addr1	MAPM ADDR1
10	MAPM_Addr2	MAPM ADDR2
11	MAPM_Addr3	MAPM ADDR3
12	Payload_size	payload长度
...	PAYLOAD	RF TX Payload DATA
...		
...		
	CRC_L	CRC16低8位(上面所有字节)
	CRC_H	CRC16高8位

Buffer size

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(3) The master controller configures parameters for the gateway module using the following protocol format:

参数配置指令	数据存储	说明
数据段起始地址		
0	CMD_CONFIG(0x0A)	参数配置区域
1	buffer_size[0]	整个数据包的长度低字节
2	buffer_size[1]	整个数据包的长度高字节
3	p_center_freq	前级中频（参考值1MHz）（默认值1000000）
4		
5		
6		
7	filter_bw	滤波带宽(0:500K, 1:1M, 2:2M)（默认值2）
8	AGC_ENABLE	AGC是否开启1：开启，0：关闭（默认值0）
9	AGC_REG1	固定增益1（默认值0xde）
10	AGC_REG2	固定增益2（默认值0xf9）
11	AGC_REG3	固定增益3（0x3c）
12	SF0[7:4]CR0[3:1]CRC_EN0[0]	通道0接收参数 SF:5-12, 智能搜索（0:SF5-8, 1:6-9, 2:7-10, 3:8-11, 4:9-12） CR:1-4, CRC:0-1 （默认值0x79）
13	BW0[7:4]LDR0[0]	（默认值0x70）
14	FREQ0	通道0频点，单位：Hz，低字节在前（默认值484500000）
15		
16		
17		
18	SF1[7:4]CR1[3:1]CRC_EN1[0]	（默认值0x79）
19	BW1[7:4]LDR1[0]	（默认值0x70）
20	FREQ1	通道1频点，单位：Hz，低字节在前（默认值484200000）
21		
22		
23		
24	SF2[7:4]CR2[3:1]CRC_EN2[0]	（默认值0x79）
25	BW2[7:4]LDR2[0]	（默认值0x70）
26	FREQ2	通道2频点，单位：Hz，低字节在前（默认值483900000）
27		
28		
29		
30	SF3[7:4]CR3[3:1]CRC_EN3[0]	（默认值0x79）
31	BW3[7:4]LDR3[0]	（默认值0x70）
32	FREQ3	通道3频点，单位：Hz，低字节在前（默认值483600000）
33		
34		
35		
36	SF4[7:4]CR4[3:1]CRC_EN4[0]	（默认值0x79）
37	BW4[7:4]LDR4[0]	（默认值0x70）
38	FREQ4	通道4频点，单位：Hz，低字节在前（默认值483300000）
39		
40		
41		
42	SF5[7:4]CR5[3:1]CRC_EN5[0]	（默认值0x79）
43	BW5[7:4]LDR5[0]	（默认值0x70）
44	FREQ5	通道5频点，单位：Hz，低字节在前（默认值483000000）
45		
46		
47		
48	SF6[7:4]CR6[3:1]CRC_EN6[0]	（默认值0x79）
49	BW6[7:4]LDR6[0]	（默认值0x70）
50	FREQ6	通道6频点，单位：Hz，低字节在前（默认值482700000）
51		
52		
53		
54	SF7[7:4]CR7[3:1]CRC_EN7[0]	（默认值0x79）
55	BW7[7:4]LDR7[0]	（默认值0x70）
56	FREQ7	通道7频点，单位：Hz，低字节在前（默认值482400000）
57		
58		
59		
60	CRC_L	CRC16低8位(上面所有字节)
61	CRC_H	CRC16高8位

Buffer size

Revision History

Version	Date	Content
1.0	2023.09	First edition

Documentation

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